

EE708 Term Project

ShiftGuard-10: Robust Image Classification Challenge

GROUP-18

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Problem Statement :

Develop a robust image classification model that generalizes well under distribution shift to accurately classify 32×32 RGB images into 10 classes.

Dataset Constraints

- **Low Resolution:** Images are limited to 32×32 pixels, providing very little spatial detail for the model to learn meaningful features.
- **Severe Class Imbalance:** The dataset has uneven representation across the 10 classes, which can bias the model toward more frequent categories.
- **Distribution Shift:** The test data contains unseen variations and distortions that differ from the relatively clean training data.

The Starting Point: Understanding the Data

- **First Look:** The dataset consists of 32×32 RGB images across 10 classes, with limited spatial detail and inherent noise.
- **The Challenge:** Low resolution, class imbalance, and distribution shift make it difficult to learn stable and generalizable features.
- **The Goal:** Before optimizing the model, it was essential to understand data characteristics and identify potential sources of bias.
- **The Strategy:** Analyze class distribution, apply stratified splitting, and design preprocessing steps to ensure balanced and robust learning.

Model Architecture

PHASE 1 :Establishing the Baseline & Identifying Limitations

Baseline: WideResNet (Initial Setup)

- **Architecture:** A WideResNet (WRN-28-10) model was used as the base architecture for feature extraction.
- **Training Setup:** Trained using standard preprocessing (normalization) with a simple cross-entropy loss.
- **Complexity:** High-capacity model, but initially trained without advanced regularization or balancing techniques.
- **Result:** Achieved reasonable training performance but showed instability and bias across classes.
- **Observation:** The model was sensitive to class imbalance and prone to overfitting on dominant classes.

PHASE 2 : Addressing Imbalance & Improving Learning Stability

Improvement: Class Balancing + Loss Adjustment

- **Sampling Strategy:** Introduced weighted sampling to ensure fair representation of minority classes during training.
- **Loss Function:** Applied adjusted cross-entropy with class frequency correction.
Objective: Reduce bias toward majority classes and improve macro-level performance.
- **The Result:** Improved balance in predictions across classes and better validation F1 score.
- **Observation:** While class-wise performance improved, the model still struggled with robustness and unseen variations.

PHASE 3: Building Robustness Through Augmentation

Tackling Overfitting & Data Limitations

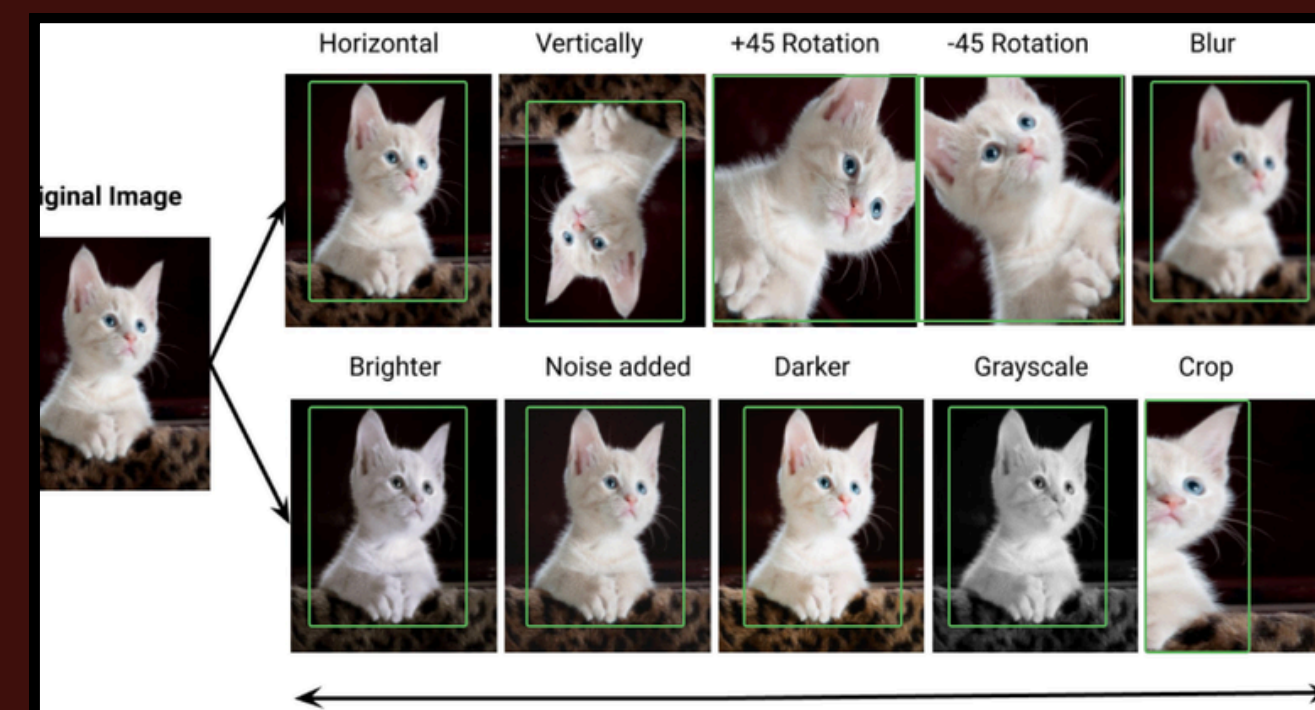
- **Augmentation Stack:** Applied Random Crop, Horizontal Flip, and AutoAugment.

- **Occlusion Handling:** Introduced Patch Masking to simulate missing or corrupted regions.

- **Objective:** Force the model to learn robust and invariant features, not surface-level patterns.

- **The Result:** Improved generalization and reduced sensitivity to small perturbations.

- **Observation:** Basic augmentation helped, but the model still struggled with complex variations and distribution shift.



PHASE 4 : Advanced Regularization & Feature Mixing

Making the Model More Generalizable

- **Data Mixing Techniques:**

MixUp: Blending images and labels

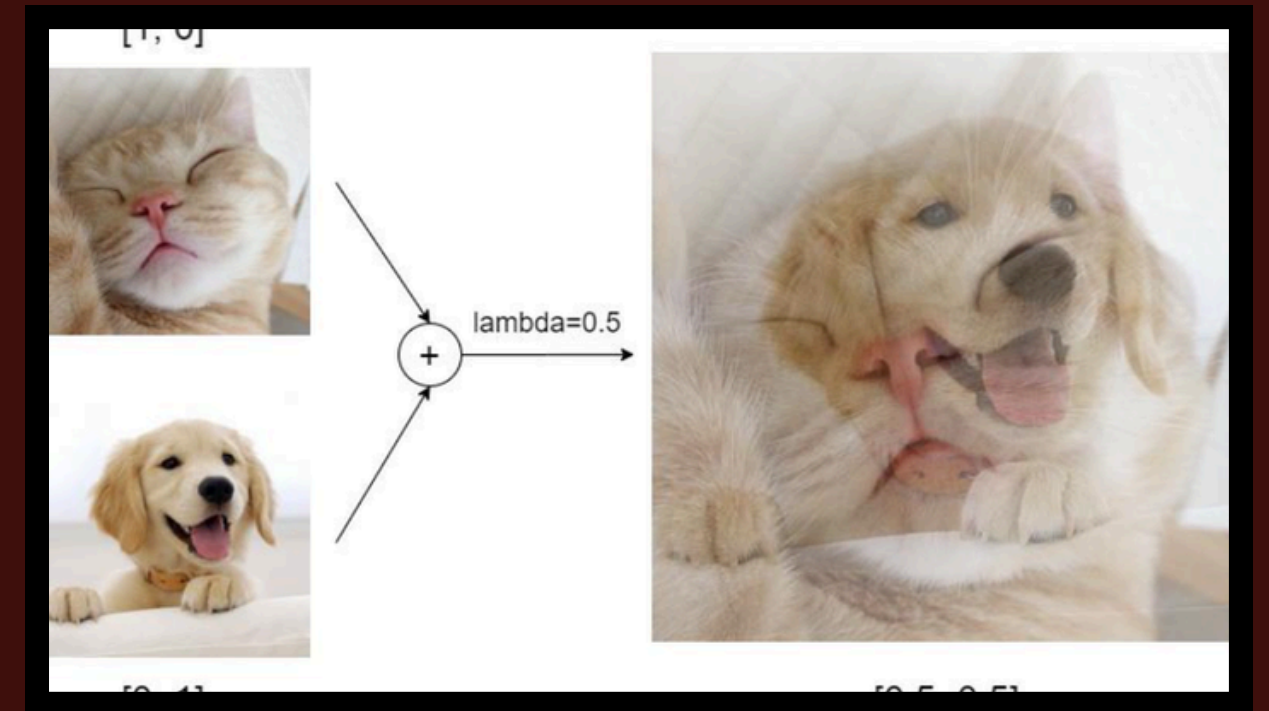
CutMix: Replacing patches between images

- **Label Smoothing:** Reduced model overconfidence and improved calibration.

- **Goal:** Encourage the model to learn soft decision boundaries instead of memorizing data.

- **The Result:** Significant improvement in validation stability and Macro F1.

Insight: The model became more robust to noise and unseen patterns.



PHASE 5: Optimization & Training

Improving Convergence and Performance

- **Optimizer:** Used AdamW with weight decay for better generalization.

- **Learning Strategy:** Applied warm-up + cosine learning rate scheduling.

```
if ep < wu:  
    return (ep + 1) / wu
```

```
t = (ep - wu) / (cfg.epochs - wu)  
return 0.5 * (1 + math.cos(math.pi * t))
```

- **Stability Measures:** Gradient clipping to prevent exploding gradients.

- **Objective:** Achieve smoother and more stable training dynamics.

- **The Result:** Faster convergence and improved overall performance.

PHASE 6 : Model Averaging for Generalization

Leveraging Stochastic Weight Averaging (SWA)

- **SWA Implementation:** Averaged model weights over final epochs.

$$W_{swa} = (W_{95} + W_{96} + \dots + W_{109}) / N$$

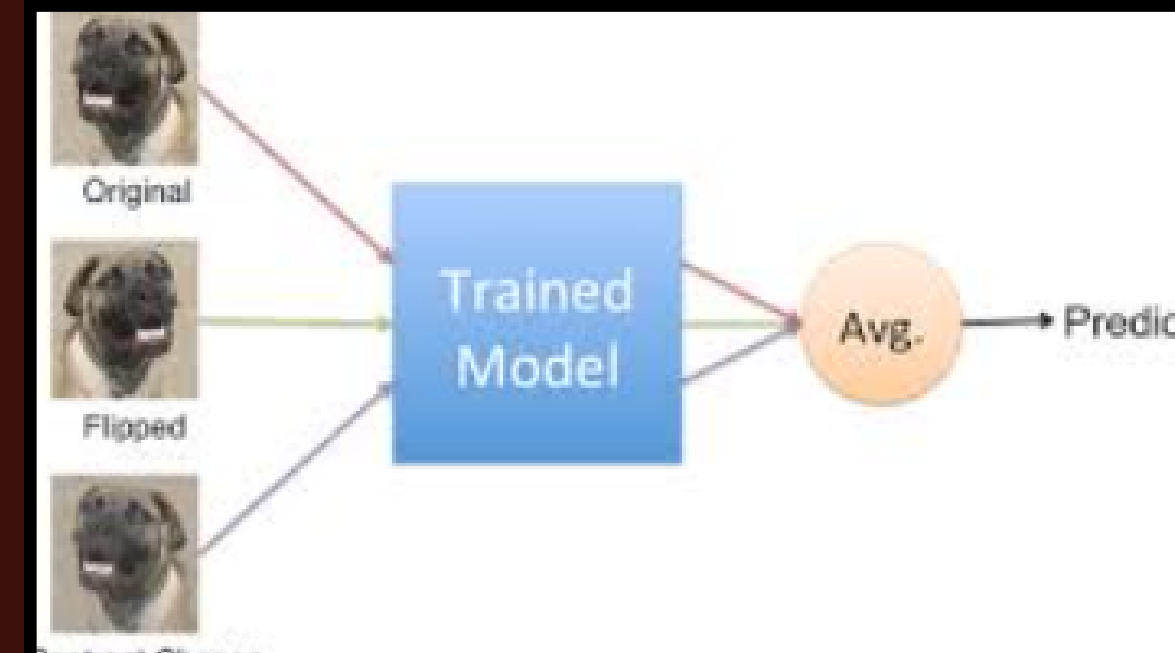
- **Why SWA?** Moves to flat minimum instead of sharp minimum



- **BatchNorm Update:** Recomputed statistics for improved inference.

- Boost in validation F1 and more stable predictions.

To generate the final predictions, we improved robustness and reduced variance using SWA and Test-Time Augmentation:



Results and Performance

Our training pipeline showed strong and consistent validation performance:

Best Non-SWA F1: 0.851 (Epoch 91)

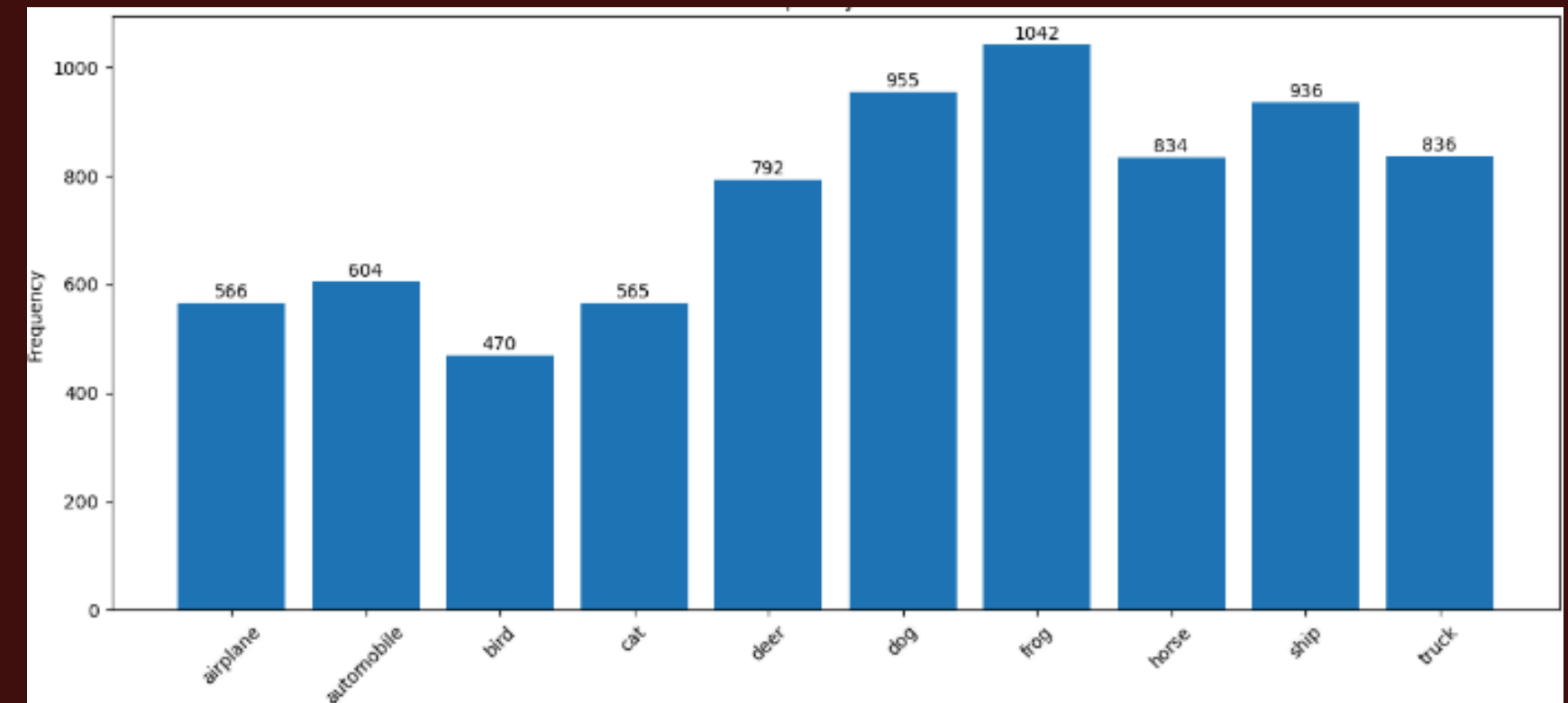
Best Epoch F1: 0.870 (Epoch 97)

Final SWA F1: 0.8808 (Best Overall)

Approx Validation Accuracy: ~91%

Prediction Distribution Highlights:

- Frog: 1042 (Most predicted)
- Dog: 955
- Bird: 470 (Least predicted)



Our validation results show steady and consistent improvement throughout training, with **Macro F1** increasing from **0.173** to **0.8808**. The use of cosine learning rate, strong augmentations, and SWA led to stable convergence and better generalisation.

Key Takeaways & Conclusion

Key Takeaways:

- Architecture: WideResNet-28-10 captures richer features than deeper, thinner models, ensuring better generalization for 32x32 images.
- Augmentation: Combining Mixup, CutMix, and AutoAugment smooths decision boundaries, preventing overfitting and boosting Macro F1-score.
- Stability: SWA (Stochastic Weight Averaging) finds flatter loss minima, while TTA reduces prediction noise for robust final results.

Conclusion:

To conclude, robust image classification requires more than just a deep network. By combining **WideResNet** with strong augmentations and stabilisation techniques like **SWA** and **TTA**, we built a highly reliable and generalisable classifier, achieving a **Macro F1-score** of **0.8808**.